Rec / Trans.

i is rec. iff $f_{ii}^{*} = P(X_n = i$ for some $n \geq 1 \mid X_0 = i)$
$= 1.$

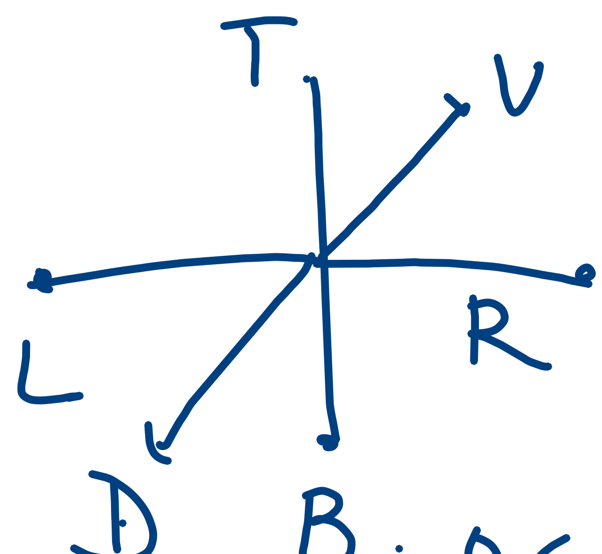
Thm:- i is rec. iff $\boxed{\sum_{n=0}^{\infty} p_{ii}^{(n)} = \infty}$.

- Simple sym. RW is rec, but asym. RW is transient.
- Two dimensional simple sym RW is rec.

__Ex__ : 3 dimensional simple sym.

RW.

$$p^{(n)}_{(0,0,0),(0,0,0)} = \begin{cases} 0 & \text{if } n \text{ odd.} \\ & n = 2m. \end{cases}$$



$$\sum_{\substack{0 \le l,t \\ l+t \le m}} \frac{(2m)!}{l!\,l!\,t!\,t!\,u!\,u!} \cdot \frac{1}{(6)^{2m}}$$

$$u!\,u! \to (m-l-t)!\,(m-l-t)!$$

$$p^{(2m)}_{0,0} = \sum_{\substack{0 \le l,t \\ l+t \le m}} \frac{(2m)!}{\left(l!\,t!\,(m-l-t)!\right)^2} \cdot \frac{1}{6^{2m}}$$

$$= \underline{\binom{2m}{m}!\ \frac{1}{6^{2m}} \cdot 3^{2m}} \sum_{\substack{0 \le l,t \\ l+t \le m}} \underbrace{\left(\frac{m!}{l!\,t!\,(m-l-t)!} \frac{1}{3^m} \right)^2}_{= \left(\alpha_{l,t}\right)^2}$$

$$\alpha_{l,t} = \frac{m!}{l!\,t!\,(m-l-t)!} \frac{1}{3^m}$$

$$= P\left(M_{m,\left(\frac{1}{3},\frac{1}{3},\frac{1}{3}\right)} = (l, t, m-l-t) \right)$$

$$\sum_{\substack{0 \le l,t \\ l+t \le m}} \alpha_{l,t} = 1$$ (total prob for the multinomial distr^n)

$$p^{(2m)}_{0,0} = \sum_{\substack{0 \le l,t, \\ l+t \le m}} (\alpha_{l,t})^2 \le \sum_{\substack{0 \le l,t \\ l+t \le m}} \left(\max_{\substack{0 \le l,t \\ l+t \le m}} \alpha_{l,t} \right) \alpha_{l,t}.$$

$$\binom{2m}{m} \frac{1}{2^{2m}}. = \max_{\substack{0 \le l,t, \\ l+t \le m}} \alpha_{l,t}.$$

Show that

$$\alpha_{l,t} \leq \begin{cases} \alpha_{k,k} & \text{if } m = 3k. \\ \alpha_{k+1,k} & \text{if } m = 3k+1 \\ \alpha_{k+1,k+1}, & \text{if } m = 3k+2. \end{cases}$$

Case $m = 3k$

$$P^{(6k)}_{\underline{0},\underline{0}} \leq \binom{6k}{3k} \frac{1}{2^{6k}} \cdot \alpha_{k,k}.$$

$$= \frac{(6k)!}{(3k)!\,(3k)!} \cdot \frac{1}{2^{6k}} \cdot \frac{(3k)!}{k!\,k!\,k!} \cdot \frac{1}{3^{3k}}.$$

$$\sim \frac{\sqrt{\pi 6k}\left(\frac{6k}{e}\right)^{6k}}{\left(\sqrt{\pi 3k}\left(\frac{3k}{e}\right)^{3k}\right)^2} \cdot \frac{1}{2^{6k}} \cdot \frac{\sqrt{\pi 3k}\left(\frac{3k}{e}\right)^{3k}}{\left(\sqrt{\pi k}\left(\frac{k}{e}\right)^{k}\right)^3} \cdot \frac{1}{3^{3k}}$$

$$= \frac{1}{\pi^{3/2}} \times C \times \frac{1}{k^{3/2}}.$$

$$p^{(6k)}_{\underline{0},\underline{0}} \leq \frac{C_1}{k^{3/2}} . \quad \Bigg| \quad p^{(6k+2)}_{\underline{0},\underline{0}} \leq \frac{C_2}{k^{3/2}}, \quad p^{(6k+4)}_{\underline{0},\underline{0}} \leq \frac{C_3}{k^{3/2}}.$$

Thus,

$$p_{\underline{0},\underline{0}}^{(6k+r)} \leq \frac{C}{k^{3/2}}, \quad r=0,2,4.$$

$$\sum_{n=0}^{\infty} p_{\underline{0},\underline{0}}^{(n)} = \sum_{m=0}^{\infty} p_{\underline{0},\underline{0}}^{(2m)} = \sum_{k=0}^{\infty} p_{\underline{0},\underline{0}}^{(6k)} + \sum_{k=0}^{\infty} p_{\underline{0},\underline{0}}^{(6k+2)} + \sum_{k=0}^{\infty} p_{\underline{0},\underline{0}}^{(6k+4)} < \infty$$

$\Rightarrow$ '$\underline{0}$' is transient.

Class :-

We define a relation between states of a chain.

We say $i \rightsquigarrow j$ if $p_{ij}^{(n)} > 0$ for some $n \geq 0$
(i leads to j)

and say i communicates with j if $i \rightsquigarrow j$ & $j \rightsquigarrow i$.

- Communication is a equivalence relation.

- Reflexive $\Rightarrow$ By def.
- Sym $\Rightarrow$ i comm. with i

$\Downarrow$

$$p_{ii}^{(0)} = 1 > 0.$$

Transitive. $\Rightarrow$ $\underline{i \leftrightarrow j}$ & $j \leftrightarrow k$

$\Rightarrow$ $i \leftrightarrow k$.

$i \leftrightarrow j$ $\Rightarrow$ $\exists\, n_1, m_1 \geq 0$, s.t

$p_{ij}^{(n_1)} > 0$, $p_{ji}^{(m_1)} > 0$.

$j \leftrightarrow k$ $\Rightarrow$ $\exists\, n_2, m_2 \geq 0$ s.t.

$p_{jk}^{(n_2)} > 0$, $p_{kj}^{(m_2)} > 0$.

$$p_{ik}^{(n_1+n_2)}$$

$$\geq p_{ij}^{(n_1)} p_{jk}^{(n_2)}$$

$$> 0 \Rightarrow i \leftrightarrow k.$$

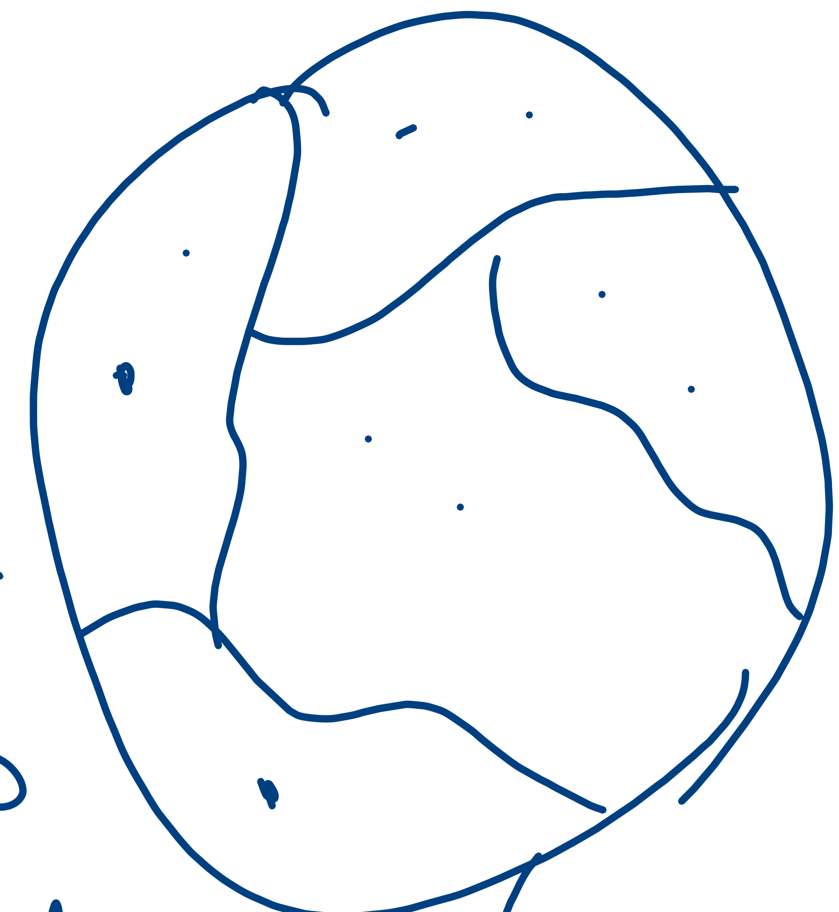
$\boxed{\text{CK eq}^n}$

Similarly, $p_{ki}^{(m_2+m_1)} \geq p_{kj}^{(m_2)} \, p_{ji}^{(m_1)} > 0.$

$\Downarrow$

$k \to i$

Thus, $i \leftrightarrow k$.



The equivalence relation partitions the state space into disjt. sets, called equivalence (communication classes) classes, such that for any two ~~s~~tates $i, j \in C$, comm. class $i \leftrightarrow j$ but for $i \in C_1$, $j \in C_2$, $C_1 \neq C_2$, either $i \not\to j$ or $j \not\to i$ or both.

<u>Thm</u> Rec/transience is a class property.

Thm: Pos rec/null rec. is a class pr.

Ex :-

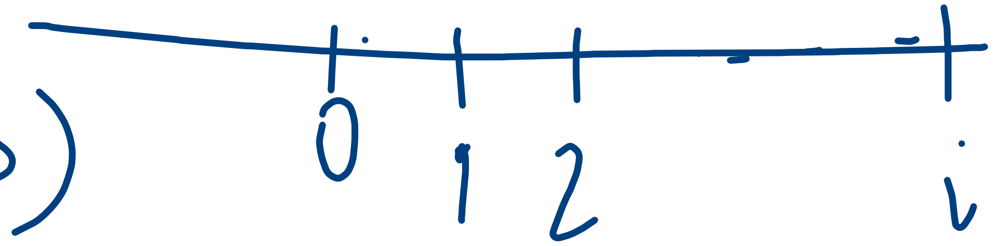
$S = \{1, 2\}$. $P = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$. $\begin{bmatrix} 1 & 0 \\ 1/2 & 1/2 \end{bmatrix}$

- $p_{12} = 0$, then $1 \not\rightsquigarrow 2$.
- $p_{21} > 0$, then $2 \rightsquigarrow 1$. $(n = 1)$

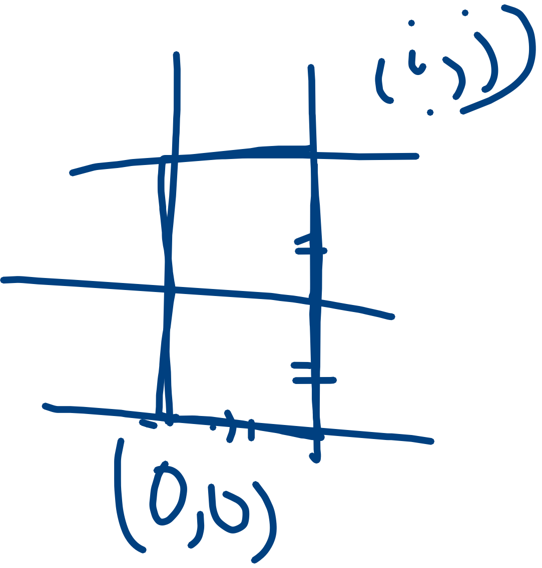
$C_1 = \{1\}$, $C_2 = \{2\}$.

If $p_{12} > 0$, $p_{21} > 0$, $C_1 = \{1, 2\}$.

RW (1-d dim).



$0 \to i \ (i>0), \ (i<0)$



$p_{0i}^{(i)} \geq p^{i} > 0$, show $i \to 0$.

$p_{0,i}^{(-i)} \geq q^{-i} > 0$

All states communicate with each other. $C = S$.

Def :- If a chain has only one comm. class, it is called irreducible.

Ex:- $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$

$$P = \begin{array}{c|cccccccc|} & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \hline 0 & * & 0 & 0 & * & & & & \\ 1 & * & 0 & * & & 0 & 0 & * & 0 \\ 2 & & & * & & * & & & * \\ 3 & * & & & * & & & & \\ 4 & & & * & & * & & & \\ 5 & * & * & * & * & * & * & * & * \\ 6 & * & & & & * & & & * \\ 7 & & & * & & * & & & * \end{array}$$

$C_1 = \{0, 3\}$. ✓ Rec

$C_2 = \{2, 4, 7\}$ ✓

$$\left.\begin{array}{l} C_3 = \{1\}. \\ C_4 = \{6\}. \\ C_5 = \{5\}. \end{array}\right\} T \text{ tran}$$

Gambler's Ruin

$$P = \begin{array}{c|c} & \begin{matrix} 0 & 1 & 2 & \cdots & N \end{matrix} \\ \hline \begin{matrix} 0 \\ 1 \\ \vdots \\ \\ N \end{matrix} & \begin{pmatrix} 1 & 0 & & & \\ q & 0 & p & & 0 \\ & \ddots & \ddots & \ddots & \\ 0 & & q & 0 & p \\ 0 & \cdots & & 0 & 1 \end{pmatrix} \end{array}$$

$C_1 = \{0\}$, $C_2 = \{N\}$ $\to$ Rec.

$C_3 = \{1, 2, \ldots, N-1\}$ $\to$ Tr.

$1 \to 0$ but $0 \not\to 1$